

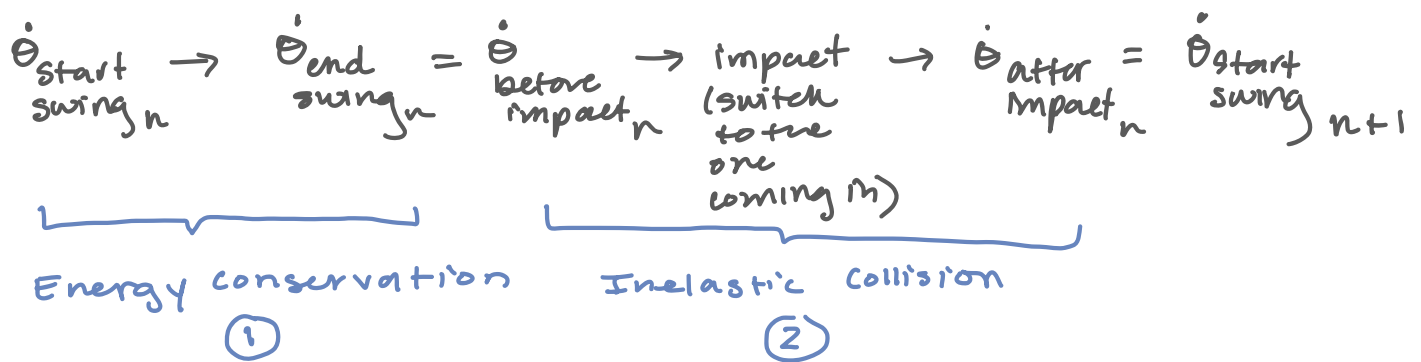
Determine Floquet Multiplier

↳ To determine the Floquet multiplier we need to take derivative of $\dot{\theta}_{n+1} = f(\dot{\theta}_n)$ or $\dot{\theta}_{n+1} = Z \cdot \dot{\theta}_n$

$$\frac{d \dot{\theta}_{n+1}}{d \dot{\theta}_n} = Z \quad \text{how } \dot{\theta}_{n+1} \text{ changes as a function of } \dot{\theta}_n$$

Essentially, given the angular velocity at the start of a swing $\dot{\theta}_{\text{start } n}$, what is $\dot{\theta}_{\text{start } n+1}$? Then, how does that change if $\dot{\theta}_{\text{start } n}$ is perturbed?

↳ first derive $\dot{\theta}_{\text{start swing } n+1} = f(\dot{\theta}_{\text{start swing } n})$



↳ (1) solve for $\dot{\theta}_{\text{end swing } n} = f(\dot{\theta}_{\text{start swing } n})$ w/ energy conservation

$$E_{\text{start}} = \frac{1}{2} \dot{\theta}_{\text{start}}^2 + \frac{g}{l} \cos \theta_{\text{start}}$$

$$E_{\text{end}} = \frac{1}{2} \dot{\theta}_{\text{end}}^2 + \frac{g}{l} \cos \theta_{\text{end}}$$

$$\dot{\theta}_{\text{end } n}^2 = \dot{\theta}_{\text{start } n}^2 + \frac{2g}{l} (\cos \theta_{\text{start } n} - \cos \theta_{\text{end } n})$$

↳ (2) Inelastic collision angular velocity change

$$\dot{\theta}_{\text{before impact } n} = \dot{\theta}_{\text{after impact } n} \cdot \cos(2\alpha)$$

↳ Equate the two and solve for $\dot{\theta}_{\text{start swing}_{n+1}} = f(\dot{\theta}_{\text{start swing}_n})$

$$\dot{\theta}_{\text{after impact}_n} \cdot \cos(2\alpha) = \sqrt{\dot{\theta}_{\text{start}_n}^2 + \frac{2g}{l} (\cos \theta_{\text{st}_n} - \cos \theta_{\text{end}_n})}$$

$$\dot{\theta}_{\text{start swing}_{n+1}} = \cos(2\alpha) \sqrt{\dot{\theta}_{\text{start}_n}^2 + \frac{2g}{l} (\cos \theta_{\text{st}_n} - \cos \theta_{\text{end}_n})}$$

where $\theta_{\text{st}_n} = \gamma - \alpha$ and $\theta_{\text{end}_n} = \gamma + \alpha$

↳ take derivative to solve for Floquet multiplier

↳ use substitution to take derivative

$$\frac{d}{dx} \left(y = a \sqrt{x^2 + C} \right)$$

$$\frac{dy}{dx} = \frac{1}{2} a (x^2 + C)^{-1/2} \cdot 2x$$

$$y = a \sqrt{x^2 + C}$$

but at the fixed point
 $y = x$

$$x = a \sqrt{x^2 + C}$$

$$\frac{x}{a} = \sqrt{x^2 + C}$$

$$\frac{a}{x} = (x^2 + C)^{-1/2}$$

↳ substitute fixed point

$$\frac{dy}{dx} = \frac{1}{2} a \cdot \frac{a}{x} \cdot 2x$$

$$\frac{dy}{dx} = a^2 = (\cos(2\alpha))^2$$

$$\boxed{\frac{d\theta_{n+1}}{d\theta_n} = Z = (\cos(2\alpha))^2}$$

The Floquet multiplier is dependent on only α so it will not change as a function of γ . It is a function of N since $a = f(N)$